

FEASIBILITY STUDY

BodyBalance AI – AI-Powered Health, Fitness, and Recovery Decision-Support Platform

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1. General Information

Business Activity Name

BodyBalance AI

Business Activity Owner's Name

Leen Kasem, Reema Abualnaja, and Farah Shetawi

Business Activity Idea

BodyBalance AI is an AI-powered health, fitness, and recovery decision-support platform designed to help users understand their current physical and mental condition and determine an appropriate next step.

The platform collects information through a personal profile and a daily physical and mental check-in. Users can report factors such as pain, fatigue, energy level, stress, sleep, recovery status, symptoms, and recent physical activity. When pain is present, the Pain Locator allows the user to identify the affected area and provide additional details such as pain intensity, when it started, whether it followed an activity or injury, and how it has changed over time.

When visual information may be useful, users can optionally use BodyScope AI, the computer-vision component of BodyBalance AI. Depending on the situation, the user may upload a close-up image of an affected area, a wider image showing the surrounding area, or a full-body/posture image. BodyScope AI considers supported visible physical indicators together with the user's reported symptoms, activity information, and available history.

Using the collected information, BodyBalance AI provides personalized activity and recovery guidance. Depending on the user's condition, the system may recommend maintaining normal activity, modifying a workout, reducing intensity, resting, monitoring the condition, or considering professional evaluation.

The platform also maintains a progress history so that users can track whether pain, discomfort, mobility, or other reported symptoms have improved, remained unchanged, or worsened over time.

BodyBalance AI is designed as a decision-support and wellness platform rather than a replacement for healthcare professionals. AI-generated information and BodyScope observations do not represent a definitive medical diagnosis or treatment decision. Medical diagnosis and treatment remain the responsibility of qualified healthcare professionals.

Reason for Choosing the Business Activity

The idea was selected because people may have difficulty deciding what physical activity is appropriate when their fitness goals conflict with their current physical or mental condition.

For example, a person may plan to exercise while experiencing poor sleep, high stress, fatigue, recurring muscle pain, bruising, discomfort, or symptoms following a previous training session. A traditional fitness recommendation based mainly on a long-term fitness goal may not sufficiently consider these temporary conditions.

BodyBalance AI addresses this problem by combining mental state, physical condition, symptoms, activity information, recovery history, and optional visual information into one decision-support process rather than treating each factor separately.

The platform therefore aims to answer a practical question:

“Considering how I feel and what is happening with my body today, what is an appropriate next step?”

The project's previous market study identified an opportunity between general fitness applications and more specialized healthcare and rehabilitation platforms. BodyBalance AI aims to address this opportunity by integrating physical and mental check-ins, symptoms, BodyScope computer vision, personalized activity and recovery guidance, progress tracking, and professional referral support within one platform.

Legal Form of the Business Activity

Company.

The final legal structure would be determined before commercial establishment according to applicable requirements and the needs of the founders.

Partners and Roles

Partner	Proposed Responsibility	Proposed Ownership
Leen Kasem	Product Development, Software Development, AI, and Project Management	33.3%
Reema Abualnaja	Product Development, Software Development, AI, and Project Management	33.3%
Farah Shetawi	Marketing, Documentation, and Administrative Support	33.3%

The proposed roles reflect the initial division of responsibilities among the team members and may be adjusted as the project develops.

2. Executive Summary

BodyBalance AI is a proposed digital health, fitness, and recovery decision-support platform that combines artificial intelligence, computer vision, user-reported information, and progress tracking to help individuals make more informed decisions about physical activity and recovery.

The platform addresses a gap between general fitness applications and specialized healthcare or rehabilitation systems. Many fitness applications focus mainly on goals such as performance, weight management, workout planning, or activity tracking, while healthcare and rehabilitation platforms generally focus on specific clinical or recovery needs. BodyBalance AI aims to connect these areas by considering the user's current condition before recommending an action.

The user journey begins with secure registration and profile creation. Users can then complete physical and mental check-ins, identify pain through the Pain Locator, optionally upload images for BodyScope AI analysis, receive personalized activity or recovery guidance, and track changes through follow-up assessments.

The three principal user groups are active students and fitness users, fitness coaches, and healthcare professionals.

Active students and fitness users use the platform to assess their current condition, receive activity or recovery recommendations, and track their progress.

Fitness coaches, with the client's permission, can review relevant fitness, activity, and recovery information to support safer and more personalized training decisions.

Healthcare professionals, with the user's permission, can review organized symptom, activity, BodyScope, and recovery information to support their independent professional assessment. The system supports professional decision-making but does not replace clinical judgment, diagnosis, or treatment.

The broader system also contains scenarios relevant to MMA and combat-sports users, first-aid situations, older adults, posture and movement concerns, visible injuries, and sensitive injury-documentation situations. These scenarios represent opportunities for gradual product expansion rather than functions that must all be commercially launched at the same time.

The proposed business model combines free basic access with premium individual features and future professional or organizational subscriptions.

The preliminary feasibility assessment supports developing BodyBalance AI initially as a focused minimum viable product (MVP). Commercial expansion should depend on user testing, technical validation of the AI and computer-vision functions, data-security measures, professional input, and confirmation of relevant regulatory requirements.

3. Description of Products and Services

Main Product and Service

BodyBalance AI will primarily be provided through a mobile application, with the possibility of a supporting web platform for professional users.

The platform consists of interconnected services rather than a single fitness recommendation feature.

Account and Profile Management

Users can securely create an account, log in, recover a forgotten password, update relevant profile information, and securely log out.

During initial profile setup, users provide information such as age, activity level, fitness background, goals, and relevant physical limitations. This information is used to personalize future recommendations.

Physical and Mental Check-In

Users can complete a short check-in about their current physical and mental condition. The system may ask about pain, fatigue, energy, stress, sleep, recovery, symptoms, and recent physical activity.

The check-in helps the system understand the user's current condition before generating recommendations.

Pain Locator

When a user experiences pain, the Pain Locator allows them to select the affected area and provide additional information, including when the pain started, its intensity, whether it followed an activity or injury, and whether it has changed over time.

This provides the system with structured information rather than relying only on a general statement that the user is experiencing pain.

BodyScope AI

BodyScope AI is the computer-vision component of the platform.

When visual information may be useful, the user can upload a relevant image. Depending on the situation, the system may request a close-up image of an affected area, a wider image showing the surrounding area, or a full-body/posture image.

BodyScope AI considers supported visible indicators together with the user's reported symptoms and activity information. The system clearly informs users that the analysis does not replace professional medical diagnosis.

Posture and Movement Analysis

Users who are concerned about posture or movement-related patterns may provide an appropriate full-body or posture image together with information about discomfort and physical activity.

BodyScope AI can provide non-diagnostic observations regarding supported visible posture patterns and use the information when generating relevant recommendations.

Visible Injury and Bruising Documentation

Users can document visible physical changes such as bruising or swelling by uploading images and providing information about how the injury occurred, when it appeared, pain level, and related symptoms.

The system can maintain these records over time to allow users to monitor changes.

Personalized Activity Recommendation

After considering available information from the user's profile, check-in, symptoms, recent activity, and BodyScope analysis, BodyBalance AI can generate a personalized activity recommendation.

Depending on the available information, the recommendation may suggest maintaining normal activity, modifying a workout, reducing intensity, temporarily stopping certain activities, or focusing on recovery.

Personalized Recovery Plan

When recovery is appropriate, BodyBalance AI can provide a personalized recovery plan based on the user's reported condition and activity.

The user can follow the recommendations and later return to report changes in their condition.

Follow-Up and Progress Tracking

After a previous assessment, injury report, or recovery plan, users can report whether their pain, discomfort, mobility, or other symptoms have improved, remained unchanged, or worsened.

The system records these changes and creates a progress history that users can review over time.

Concerning Symptoms and Escalation

If the information provided by a user contains concerning indicators, the system does not continue with normal fitness recommendations as if the situation were routine.

Instead, BodyBalance AI can inform the user that professional or urgent medical evaluation may be appropriate.

Professional Referral Summary

When professional evaluation is recommended, BodyBalance AI can organize relevant information collected during the user's session, such as symptoms, affected area, activity history, recovery progress, and available BodyScope observations.

This summary can help the user communicate their situation more clearly when seeking professional care.

Fitness Coach Access

With the client's permission, a fitness coach or personal trainer can access relevant fitness, activity, and recovery information.

The coach can review patterns such as recurring discomfort following particular exercises and use the information to adjust the client's training plan.

Healthcare Professional Access

With the user's permission, a healthcare professional can review an organized history of reported symptoms, physical activity, recovery progress, and previous BodyScope observations.

The information supports the professional's assessment but does not replace clinical judgment, diagnosis, or treatment.

Permission and Information Sharing

Users remain in control of their information.

When information is shared with a coach or healthcare professional, the system requests the user's permission. The user can determine what information is shared and can later change or remove access.

History Review

Users can review previous check-ins, reported symptoms, BodyScope observations, recommendations, recovery plans, and progress information to better understand changes in their condition over time.

Additional Supported and Future Use Cases

As the platform develops, BodyBalance AI may support more specialized contexts.

For MMA, boxing, and combat-sports users, the system may help document and monitor training-related bruising, muscle pain, joint discomfort, and other concerns.

For first-aid situations, the system may collect information about what happened, the affected area, symptoms, and possible warning signs before providing basic guidance within its supported scope or prioritizing professional care when appropriate.

For older adults, recommendations may consider age, reported physical limitations, symptoms, activity level, and available history.

In sensitive situations involving possible violence or abuse-related injuries, the system may allow users to privately document physical injuries without requiring them to identify the cause if they do not feel safe doing so. Privacy, discreet access, and user safety would be prioritized.

Unique Characteristics and Added Value

The main value of BodyBalance AI comes from integration.

Instead of following a simple process such as:

Fitness Goal → Workout Recommendation

BodyBalance AI follows a more context-aware process:

Profile + Current Physical and Mental State + Symptoms + Activity + History + Optional BodyScope Analysis → Personalized Next Step

This allows recommendations to reflect the user's present condition rather than only their long-term fitness goal.

The platform provides added value through personalization, continuity, integration, and professional escalation.

Personalization allows recommendations to respond to the individual's current condition.

Continuity allows users to track changes rather than completing isolated assessments.

Integration allows physical, mental, activity, recovery, and visual information to be considered together.

Professional escalation helps users recognize situations in which routine fitness guidance may no longer be appropriate.

4. Technological Impact

Positive Effects of Technology

Technology is the primary enabler of BodyBalance AI.

Artificial intelligence allows the platform to process multiple types of user information and support personalized recommendations. Computer vision allows BodyScope AI to incorporate supported visual information, while databases and cloud infrastructure allow the platform to maintain user histories and monitor changes over time.

Technology also supports scalability. Because BodyBalance AI is primarily a digital service, the platform may potentially serve an increasing number of users without requiring a physical facility for every interaction.

Automated data organization may also help coaches and healthcare professionals review authorized information more efficiently.

Potential Negative Effects of Technology

The use of artificial intelligence also introduces limitations and risks.

AI recommendations may be affected by incorrect or incomplete user input, insufficient information, poor-quality images, limitations in computer-vision models, or model bias.

Users may also place excessive confidence in an automated output because it was generated by artificial intelligence.

For this reason, BodyBalance AI should communicate uncertainty, clearly define the limitations of its functions, and distinguish decision-support information from professional medical diagnosis.

Cybersecurity and privacy are also significant considerations because the platform may process personal and health-related information.

Future Technological Impact

Advances in artificial intelligence, computer vision, smartphone-camera quality, wearable devices, cloud computing, and digital-health technologies may improve the capabilities of BodyBalance AI.

Future integration with supported wearable devices could allow information such as sleep, physical activity, heart rate, and recovery-related metrics to complement user-reported information.

However, rapid technological development also means that BodyBalance AI would require continued software maintenance, cybersecurity updates, model evaluation, and feature improvement.

Using Technology to the Business's Advantage

BodyBalance AI can use technology to create a continuous feedback process:

Assessment → Recommendation → Activity or Recovery → Follow-Up → Updated Recommendation

This provides an advantage over a one-time analysis because the platform can use previous information to support future recommendations and progress tracking.

5. Marketing Strategy

BodyBalance AI's initial marketing strategy will focus on active students, gym users, recreational athletes, and fitness-conscious young adults.

These users regularly make decisions about exercise, training intensity, rest, and recovery and therefore represent an appropriate initial target audience for the platform.

Marketing Methods

Social Media Marketing

BodyBalance AI can use social media platforms such as Instagram and TikTok to introduce the platform through short educational and demonstration-based content.

Rather than advertising only technical features, the marketing can demonstrate realistic user situations.

For example:

“You planned leg day today, but your knee has been hurting since yesterday. Should you train normally, modify your workout, or focus on recovery?”

This approach communicates the practical problem that BodyBalance AI is designed to address.

University and Student Communities

University sports clubs, fitness clubs, wellness programs, campus events, and student organizations can provide direct access to the initial student segment.

Demonstrations and trial access can allow potential users to understand the service before subscribing.

Fitness Partnerships

Gyms, personal trainers, sports clubs, and combat-sports facilities can introduce BodyBalance AI to people who regularly make exercise and recovery decisions.

Partnerships with coaches may also support the future professional version of the platform.

Healthcare and Rehabilitation Partnerships

As the platform develops and appropriate validation is completed, collaboration with physiotherapists, sports-medicine professionals, and rehabilitation providers may support professional review, referral pathways, and credibility.

Freemium User Acquisition

Basic functionality can be offered without payment so that users can experience BodyBalance AI before deciding whether premium features provide enough additional value to justify a subscription.

Suitability of the Marketing Strategy

The selected marketing channels provide direct access to people who already have an interest in exercise, recovery, wellness, and physical performance.

Instead of initially marketing BodyBalance AI to the entire population, focusing on active users provides a clearer value proposition and allows the business to validate its product with a more specific customer segment.

Competitive Differentiation

BodyBalance AI will position itself as a condition-aware health, fitness, and recovery decision-support platform.

The platform does not compete only through the number of workouts, fitness metrics, or health measurements it provides.

Its differentiation comes from combining:

Mental State + Physical Condition + Symptoms + Activity + Recovery History + BodyScope Visual Information + Personalized Guidance + Progress Tracking + Professional Escalation

within one user journey.

6. Business Activity and Human Resources

Because BodyBalance AI is primarily a software-based business, the project can initially operate with a relatively small team.

First Three Months

Estimated Team: 3–4 people

Primary responsibilities include:

- product and project management;
- software development;
- UI/UX design;
- initial AI and recommendation-system development.

During this stage, the founders may perform multiple responsibilities to reduce costs.

First Six Months

Estimated Team: 5–6 people

As development becomes more advanced, additional expertise may include:

- AI and computer-vision development;
- cybersecurity and data support;
- part-time health and fitness professional consultation.

First Year

Estimated Team: 7–10 people, depending on MVP performance, funding, and user growth.

Potential roles include:

- software engineers;
- AI/computer-vision developers;
- UI/UX designer;
- cybersecurity/data specialist;
- business and operations support;
- marketing;
- customer support;
- health and fitness advisors.

Healthcare professionals do not necessarily need to become full-time employees during the initial stage. Qualified advisors or professional partnerships may be more economically appropriate during product development and validation.

7. Timeline

Main Activity	Expected Period
Finalize business and software requirements	Month 1
Develop UI/UX prototype and technical architecture	Months 1–2
Develop registration, authentication, and profiles	Months 2–3
Develop physical/mental check-in and Pain Locator	Months 3–4
Develop initial recommendation engine	Months 3–5
Develop initial BodyScope AI prototype	Months 4–7
Integrate activity and recovery recommendations	Months 6–7
Implement history, follow-up, and progress tracking	Months 7–8
Implement permission and information-sharing functions	Month 8
Conduct internal testing and technical evaluation	Months 8–9
Conduct limited MVP user pilot	Months 9–10
Improve the system based on testing and feedback	Months 10–11
Develop initial fitness/health partnerships	Months 10–12
Controlled MVP launch	Month 12

Development Approach

The MVP will prioritize the most commercially and technically manageable user journey:

Account → Profile → Check-In → Pain Locator → Limited BodyScope → Personalized Recommendation → Recovery → Follow-Up → Escalation or Referral

The platform's more specialized scenarios will be expanded gradually after the core system has been tested and validated.

8. Technical Section

Business Location

BodyBalance AI is primarily a digital service and therefore does not require a customer-facing physical facility during the initial stage.

The business may initially operate through a university or startup workspace, shared office, or small development office in Saudi Arabia.

Suitability of the Location

Because the product is digitally delivered, the physical location of the development team does not significantly limit customer access.

Initial testing may be conducted through university communities, gyms, fitness clubs, coaches, athletes, and other active users.

A digital-first structure also reduces the need for significant physical infrastructure during the early development stage.

Required Materials and Resources

The project will primarily require:

- development computers;
- smartphones for application testing;
- secure cloud infrastructure;
- databases;
- software-development tools;
- AI and computer-vision infrastructure;
- suitable development and testing datasets;
- cybersecurity and authentication systems;
- analytics and testing tools;
- backup and recovery systems.

Operational Technical Components

The proposed technical architecture includes the following major components:

1. Authentication Module

Supports registration, secure login, password recovery, logout, and account protection.

2. User Profile Module

Stores relevant information such as activity level, fitness background, goals, and editable profile information.

3. Physical and Mental Assessment Module

Collects information regarding pain, fatigue, energy, stress, sleep, recovery, symptoms, and recent physical activity.

4. Pain Locator

Records pain location and related information such as intensity, onset, possible cause, and changes over time.

5. BodyScope Computer-Vision Engine

Processes supported images and provides non-diagnostic observations regarding available visible indicators.

6. Recommendation Engine

Combines available user information to provide personalized activity and recovery recommendations.

7. Risk and Escalation Logic

Identifies situations in which normal fitness guidance may not be appropriate and professional evaluation should be considered.

8. Recovery and Progress Module

Stores previous assessments, symptoms, recommendations, recovery plans, and progress information.

9. Professional Access Module

Allows authorized coaches and healthcare professionals to access approved information.

10. Privacy and Permission Management

Allows users to determine what information may be shared, with whom it may be shared, and when access should be removed.

Government and Regulatory Considerations

Because BodyBalance AI may process personal and health-related information, applicable requirements related to privacy, consent, cybersecurity, data protection, digital services, and potentially digital-health software would need to be confirmed before commercial deployment.

More advanced medical functionality may require additional professional validation or regulatory consideration.

For the initial feasibility stage, BodyBalance AI will therefore remain positioned as a health, fitness, and recovery decision-support platform. Medical diagnosis and treatment remain outside the intended role of the platform and under the responsibility of qualified healthcare professionals.

9. Business Model

Service Presentation

BodyBalance AI will primarily operate as a mobile application.

A supporting web-based professional interface may later be developed for authorized fitness coaches and healthcare professionals.

Distribution

The consumer application can be distributed digitally through major mobile application stores.

Additional distribution may occur through partnerships with:

- universities;
- gyms;
- personal trainers;
- sports clubs;
- combat-sports facilities;
- rehabilitation providers;
- healthcare organizations.

Pricing Model

A freemium subscription model is proposed.

Free Version

The free version may include:

- account and profile creation;
- daily physical and mental check-ins;
- basic Pain Locator;
- limited history;
- basic activity and recovery recommendations.

Premium Version

The premium version may include:

- expanded BodyScope AI functionality;
- detailed personalized recovery plans;
- extended progress tracking;
- longer history;
- advanced personalization;
- additional future integrations.

Professional and Organizational Plans

Professional subscriptions may later be offered to fitness coaches and personal trainers.

Organizational plans may be developed for gyms, sports clubs, rehabilitation providers, universities, and other appropriate organizations.

Exact commercial pricing should be finalized after MVP testing and willingness-to-pay validation rather than assuming that an untested price represents actual customer demand.

Revenue Sources

Potential revenue sources include:

- individual premium subscriptions;
- professional subscriptions;
- organizational subscriptions;
- institutional licensing;
- strategic partnerships.

Value Beyond Financial Profit

BodyBalance AI may help users become more aware of their current physical condition, make more informed exercise decisions, monitor recovery more consistently, and communicate organized information to professionals.

The platform may also contribute to bridging the gap between everyday fitness decisions and situations in which professional evaluation may become appropriate.

10. General Risk Factors

Risk	Potential Impact	Risk-Reduction Strategy
AI recommendation inaccuracy	Inappropriate activity or recovery guidance	Conservative recommendation logic, testing, validation, and escalation
BodyScope computer-vision error	Incorrect visual observations	Image-quality checks, limited supported uses, testing, and clear non-diagnostic wording
Poor-quality images	Reduced analysis reliability	Image-quality validation and guided image capture
Incorrect or incomplete user information	Less appropriate recommendations	Structured questions, follow-up inputs, and editable information
Concerning symptoms	Routine fitness recommendation may be unsafe	Stop normal recommendation flow and prioritize professional evaluation
Data/privacy breach	Exposure of sensitive information	Encryption, secure authentication, access control, and minimum necessary data collection
Unauthorized information sharing	Privacy violation and loss of trust	Explicit, limited, and revocable user permissions
User over-reliance on AI	Delayed professional care	Clear system limitations and professional or urgent-care escalation
Model bias	Unequal system performance	Diverse datasets and continuous performance evaluation
Sensitive injury information	Privacy and user-safety concerns	Discreet design and strict information controls
Regulatory changes	Features may require modification or approval	Early regulatory review and adaptable product scope
High AI/cloud costs	Reduced profitability	Model optimization and scalable infrastructure
Limited initial data	Reduced early AI performance	Focused MVP and gradual validated expansion
Low user adoption	Reduced revenue	User testing, focused marketing, freemium acquisition, and continuous product improvement
Strong competition	Difficulty gaining market share	Clear differentiation through integrated condition-aware decision support

11. Financial Expectations

Because BodyBalance AI has not yet commercially launched, the following financial figures are preliminary planning assumptions rather than historical financial results.

They are intended to evaluate the potential financial direction of the business and should be updated after MVP testing provides real information about user adoption, pricing, operating costs, and customer retention.

Main Cost Categories

The principal costs are expected to include:

- software development;
- AI and computer-vision development;
- cloud hosting;
- secure data storage;
- cybersecurity;
- software testing;
- professional health and fitness consultation;
- UI/UX development;
- application-store and developer fees;
- marketing;
- customer support;
- legal and regulatory consultation;
- model maintenance and validation.

Estimated Year 1 Budget	SAR
Software & AI Development	55,000
Cloud, Database & AI Services	15,000
Testing & Cybersecurity	10,000
Health/Fitness Professional Consultation	10,000
Marketing & Launch	30,000
Administrative, Legal & Other Costs	15,000
Contingency	15,000
Total Estimated Year 1 Budget	150,000

Preliminary Three-Year Financial Expectations

	Year 1	Year 2	Year 3
Estimated Revenue	SAR 120,000	SAR 400,000	SAR 850,000
Development and Service Costs	SAR 90,000	SAR 150,000	SAR 250,000
Marketing and Administrative Costs	SAR 60,000	SAR 100,000	SAR 160,000
Estimated Net Result	–SAR 30,000	SAR 150,000	SAR 440,000

These estimates assume that the founding team will perform a significant portion of the initial software, AI, product, and business development. Therefore, the first-year budget represents a lean MVP development approach rather than the cost of a fully staffed commercial healthcare technology company.

Income Expectations

During the first year, revenue is expected to remain relatively limited because the business will still be developing, testing, launching, and introducing the platform to users.

Revenue growth during Years 2 and 3 would depend primarily on:

- growth in registered users;
- conversion of free users to premium subscriptions;
- customer retention;
- professional subscriptions;
- organizational partnerships;
- successful expansion of premium functionality.

These revenue projections are therefore estimates rather than guaranteed results.

Cash-Flow Expectations

Cash outflows are expected to be relatively high during early development because software development, AI infrastructure, testing, professional consultation, cybersecurity, and launch activities occur before the business develops a stable recurring-revenue base.

As premium subscriptions and professional or organizational partnerships increase, recurring cash inflows are expected to become more significant.

Maintaining sufficient liquidity during development and the initial launch stage will therefore be important.

Break-Even Analysis

Based on the preliminary financial assumptions, BodyBalance AI is expected to operate below break-even during the first year because of development and launch expenses.

The preliminary projection indicates the possibility of moving toward break-even during Year 2 as recurring subscription and partnership revenue increases.

However, the actual break-even point should be recalculated after MVP testing provides reliable information regarding:

- subscription pricing;
- number of active users;
- premium conversion rate;
- customer retention;
- customer-acquisition cost;
- AI and cloud cost per user;
- professional and organizational revenue.

Cost and Revenue Sustainability

BodyBalance AI benefits from a digital business model in which one software platform can potentially serve an increasing number of users.

However, growth is not cost-free. Increased usage can increase cloud hosting, AI processing, data storage, cybersecurity, customer-support, and system-maintenance expenses.

Long-term financial sustainability will therefore depend on recurring subscription and organizational revenue increasing sufficiently to cover both fixed development expenses and variable costs associated with serving users.

12. Results and Recommendations

The feasibility study indicates that BodyBalance AI has a reasonable foundation for continued development as an AI-powered health, fitness, and recovery decision-support platform.

Positive Factors

One of the project's main strengths is that it addresses a recognizable user problem.

People may know their long-term fitness goals but still be uncertain about what they should do when their current physical or mental condition changes. Pain, fatigue, poor sleep, stress, bruising, discomfort, recent injury, or incomplete recovery may affect whether normal training is appropriate.

BodyBalance AI attempts to support this decision by considering several relevant factors together.

Another strength is that the platform provides a complete user journey rather than functioning as a single AI analysis tool.

The proposed system includes:

- secure account creation and management;
- personalized profiles;
- physical and mental check-ins;
- Pain Locator;
- BodyScope AI;
- posture and visible-injury analysis;
- personalized activity recommendations;
- personalized recovery plans;
- follow-up and progress tracking;
- concerning-symptom escalation;
- professional referral summaries;
- authorized coach access;
- authorized healthcare-professional access;
- permission management;
- history review.

The software-based model also provides opportunities for scalability and supports both individual and professional business models.

Challenges

The project also faces important challenges.

AI-generated recommendations must be sufficiently reliable for their intended scope. BodyScope computer-vision functions require testing and validation. Sensitive user information must be protected. Users must understand that the platform does not replace healthcare professionals.

The business must also control AI and cloud costs, establish user trust, attract and retain customers, and confirm applicable regulatory requirements before commercial deployment.

Another challenge is the breadth of the proposed system. Attempting to commercially launch every possible scenario simultaneously would increase technical complexity, cost, safety requirements, and development time.

Recommendation

The recommended approach is to proceed with a focused minimum viable product.

The initial product should prioritize:

Secure Account and Profile → Daily Physical and Mental Check-In → Pain Locator → Limited BodyScope AI → Personalized Activity and Recovery Recommendations → Progress Tracking → Concerning-Symptom Escalation → Professional Summary → Privacy and Permission Controls

Active students, gym users, and other fitness users should form the primary initial customer segment because they align closely with the core problem BodyBalance AI is designed to address.

Fitness-coach functionality can be introduced as the user base and platform develop.

More specialized applications, including expanded combat-sports functionality, older-adult adaptations, broader first-aid support, advanced professional integration, and sensitive injury-documentation scenarios, should be expanded gradually following appropriate technical testing, professional review, privacy assessment, and regulatory consideration.

Final Feasibility Conclusion

The preliminary feasibility study supports continuing BodyBalance AI to the **MVP development and validation stage**.

The project has a technically achievable software foundation, a defined target user group, a differentiated value proposition, several potential revenue sources, and opportunities for future expansion.

However, successful commercial feasibility will depend on validating three major areas.

Technical feasibility depends on whether the recommendation system and BodyScope AI can perform reliably within a clearly defined scope.

Operational feasibility depends on whether users, fitness professionals, and healthcare professionals can interact with the platform safely, conveniently, and with appropriate privacy and permission controls.

Financial feasibility depends on whether real user adoption, premium conversion, professional subscriptions, and partnerships can generate sufficient recurring revenue to cover development, AI processing, cybersecurity, maintenance, marketing, and growth costs.

For this reason, the recommended next step is not immediate large-scale commercialization. The appropriate next step is to develop and test a focused MVP, collect real user and operational data, evaluate the performance of the AI components, and then refine the technical, operational, and financial assumptions before expanding the platform.

BodyBalance AI therefore demonstrates **preliminary feasibility with a phased-development approach**, beginning with active-user fitness and recovery decision support and gradually expanding into its broader health, professional, and specialized use cases as the system is validated.